ABDULLAH AL MULQI

Robotics Engineering Student, KAIST

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EDUCATION

- KAIST, B.S. Electrical Engineering and Mechanical Engineering, Daejeon, Korea | Expected Feb 2029
- Middle East Technical University, B.S. Electrical Engineering, Ankara, Türkiye · GPA 3.55/4.0 (High Honors) | Oct 2024 – Jan 2025

EXPERIENCE

Robotics Intern & Hackathon Participant | Angel Robotics | Sep 2026 – Present

- Designed, assembled and programmed a dual-arm mobile manipulator in a team of three; the arms propel the wheeled base by pushing against the floor.
- Trained its behaviours by imitation learning on Angel Robotics' PHACT actuators and an NVIDIA Jetson AGX Orin (64 GB); integrated Orbbec depth cameras with NVIDIA cuVSLAM for exploration and mapping.
- Built the arm's URDF from CAD with exact four-bar linkage kinematics. Recognized by Angel Robotics for providing the most product feedback.

Robotics Research Intern | FAIR Lab (Field AI & Robotics), KAIST | Aug 2026 – Present

- Deployed a Rajant mesh network using Ackermann robots as mobile relays for a tracked client robot; implemented an obstacle-aware RSSI model calibrated against measured radio data.
- Integrated all robots into a Gazebo hospital simulation across mismatched ROS 2 and Gazebo versions on two NUCs (Docker); developing RSSI-driven relay motion to maintain connectivity.

PROJECTS

ROBONEXUS: Autonomous Capture-the-Flag Robot | Project Lead, KAIST MR Club | 2026 Spring – Present

- Leading a mecanum-wheeled robot that plays the first 45 s of each match autonomously; secured ₩6,000,000 in funding; advanced to the monitored development stage ahead of the March 2027 competition.
- Designed the robot in CAD and URDF, simulated in Gazebo and designed a physical arena replica; trained a YOLOv8n flag detector on 5,000+ synthetic Unity renders (mAP@50 of 0.94).
- Built the first prototype and programmed all motors and grippers; implemented OpenVINS stereo visual-inertial odometry (Kalibr-calibrated) and fused wheel odometry with it.

Autonomous Drone | [EE478] | Spring 2026

- Built, in a three-person team, a ROS autonomous flight stack on an NVIDIA Jetson Orin, integrating SVO, depth mapping and EKF sensor fusion to limit odometry drift.
- Implemented PID control, waypoint navigation and a YOLO-guided landing that places a waypoint above the target and descends under controlled speed.

Mobile Manipulator Rover | AI & Vision for Robotics [EE405] | Fall 2026

- Implemented ArUco detection with FK/IK grasping on a 6-DOF arm, plus SLAM mapping and Nav2 navigation (omnidirectional MPPI, A*) on a mecanum rover in ROS 2.

Autonomous Racing Model Car | Mechatronics System Design [ME209] | 2025

- Designed and built a Raspberry Pi race car combining a PyTorch CNN line follower, OpenCV start-light detection and PID control on ultrasonic range.
- Restored the complete software stack after a storage failure four days before the final demo, then repaired a motor fault to recover the project grade.

AWARDS & LEADERSHIP

- Valedictorian, Al-Rowad International Schools (2024); High Honors, Middle East Technical University (Fall 2024).
- Outstanding Cambridge Learner Award: Highest Mark in the World, AS-Level Chemistry; Highest in Saudi Arabia, A-Level Physics.
- KAIST Ambassador (2025), presenting KAIST in Saudi Arabia; Master of Ceremonies, 2024 Al-Rowad graduation (1,000+ attendees).

SKILLS

- **Robotics & AI:** ROS 2, Gazebo, Nav2, SLAM, OpenVINS, cuVSLAM, PX4, PyTorch, YOLO, OpenCV, TensorRT, imitation learning
- **Code & hardware:** Python, C, React, Docker, Linux, SolidWorks, Fusion 360, Jetson, Raspberry Pi, ESP32